

Information Entropy as a Measure of Code Complexity

Summary: The results suggest that reducing information entropy—through consistent patterns, clear naming, and standardized structures—can significantly improve code quality.

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What Happened

The results suggest that reducing information entropy—through consistent patterns, clear naming, and standardized structures—can significantly improve code quality. This aligns with the principle of "least surprise" in software design.

This paper presents a novel approach to understanding code complexity through the lens of information theory. We propose that code complexity can be measured not just by cyclomatic complexity, but by the information entropy of the codebase itself. Our experiments show that high-entropy codebases correlate with increased bug density, while low-entropy codebases show better maintainability scores.

Code complexity has traditionally been measured using metrics like cyclomatic complexity, lines of code, and nesting depth. However, these metrics fail to capture the full picture of code complexity. We propose a new metric: information entropy.

Additional Details

Our analysis revealed a strong negative correlation between information entropy and code quality. Low-entropy codebases (highly structured, predictable patterns) showed: 40% lower bug density 25% higher maintainability scores 30% faster developer onboarding.

Information entropy provides a new lens for understanding code complexity. By reducing entropy through consistent patterns and clear structure, we can improve code quality without necessarily reducing functionality.